

Answers for Collection A: Music Genre Preference Survey**Q1a:**

This dataset will be created through a survey that I create and distribute to other college students. The survey will ask participants to share their favorite music genre (from a provided list), their gender, and age. The mode of data collection is both observation and generation, as I will be generating the data through asking survey questions and observing the provided responses. I was driven to perform this survey because I wanted to perform a survey with questions that were not divisive or open ended, but rather had a select set of options for participants to choose from. I also expect to be able to use this data for a future project involving spotify playlists. The driver for the research is the question "Do music genre preferences differ by gender or age?". This survey will be administered through word of mouth conversations, text, or email, and will be collected digitally through Google Forms. Responses will be anonymous since personal information is not needed for this survey. It will also increase participant engagement in the survey. The final dataset will contain categorical variables so that the data is feasible for me to collect as a college student and usable for statistical analysis.

Q1b:

- **Creation of logical Collections**

The logical collection will consist of all the survey responses gathered over the given period of time allotted for this project. The raw data will be provided from Google Forms and will be stored as a CSV file once cleaned. Potential summaries on the data will also be stored in this manner. A clear naming convention will be used to distinguish between raw and cleaned datasets.

- **Physical data handling**

Once the data is collected via Google Forms and stored in Google Sheets, it will be downloaded as a CSV file so that it can be accessed without internet connection. The raw responses will be left unmodified, and any cleaned versions of the data will be stored in separate files. These files will be stored locally on my laptop, with backups uploaded to google drive and/or github to ensure there is a secondary backup. Should the dataset need to be transferred into a physical form, I will use a printer.

- **Interoperability Support**

The dataset will be stored in a CSV format, which is supported by many platforms such as R, Python, and Excel. Should this dataset need to be shared with collaborators, I would show all versions of the file to ensure maximum compatibility. The metadata would be stored in a plain text README file, allowing it to be read on any system. Each of these steps is essential to ensure that the data can be reused and read in different environments without compatibility issues.

- **Security Support**

While this dataset will not contain any overly sensitive personal information, it will include demographics such as gender and age. Thus, I will ensure to collect the data anonymously to protect the identities of the participants. The files will be stored in password protected accounts on my laptop, google drive and/or github. Permissions for any cloud storage would be set to "private" unless explicitly shared with a collaborator.

Finally, each time the dataset is open, I will keep it read only to further protect it against any accidental modifications.

- **Data Ownership**

Since the data will be collected via a survey of my creation, I will be the owner of the dataset. All answers will be collected voluntarily, but no participant shall retain ownership of the dataset. Should I share these files with others, I will note that the data was collected by me for educational purposes. I will maintain data quality by storing the files as read only to prevent modification.

- **Metadata collection, management, and access**

The metadata for this dataset will include its purpose, distribution method(s), collection dates, number of participants, survey questions, and variable definitions. This data will be stored in a README file which will travel with the dataset. Additional metadata will show any steps I took in processing the data, which will be stored in the same folder as the dataset. Keeping this metadata for collaborators to view is imperative to maintaining transparency and correct interpretation of the data.

- **Persistence**

The dataset along with any metadata will be backed up on my github/google drive to maintain long term persistence. The raw dataset shall be preserved without any modifications, and derived datasets will be saved with version numbers in the filenames to differentiate them. This data by default will be stored in my google drive as part of google sheets, but keeping my own copy locally ensures reproducibility in a scenario where I cannot access either google drive or github. Github will ensure that the dataset and the project itself live on past the semester and graduation. Should I want others to access it, I can explicitly share it with them or set the repository to "public".

- **Knowledge and information discovery**

Should this project be shared with others, people would be able to learn from the data without doing their own analysis by looking at the one that I store in the project folder. However, should they choose to do their own analysis they would be able to do so. I would provide insights into the relationship between different demographics and their music preferences. Additionally, discovery trends will be shown to help identify patterns and answer the research question. If I wanted to make it even easier for others to search, filter, or query the dataset, I could store it in an SQL database or provide visualizations like plots or histograms.

- **Data distribution and publication**

Once the survey result collection period is complete, I will be able to share the data as the owner of the dataset. If I were to publish my processed dataset through a means such as a github repository, I would provide my processed data, the original data, and a README file with the metadata. Any metadata or data will have potential identifiers removed before sharing to ensure that only the genre, gender, and year variables remain. This protects the anonymity of the survey participants and guarantees that it can be distributed safely for further analysis.

Q2:

The survey dataset will first exist in Google Sheets through automated generation by Google Forms. This will help make the review and collection process easier. Once the survey has concluded, the data will be exported as a CSV since many platforms support it. CSV is also a very readable format for humans wishing to interpret the data. However, it comes with storage limitations and little built in validation. JSON is also a suitable format, as it allows for structured nesting and visible metadata. Should the dataset need to be viewed from an SQL based platform, it could be converted to SQL for easy queries and joining of other data. Finally, the excel file type, XLSX, would be useful for smaller analysis or sharing information in a business context since it is typically used by less technical users. The downside for this choice is that it can be a hassle to reproduce large scale work. Overall, CSV is the most practical choice due to its large support network, but SQL would be suitable for a long term project.

Q3:

There are a few metadata standards that could be useful for this dataset. Dublin Core is widely used for dataset description, including fields such as title, creator, source, subject, and date. This schema is particularly good for inclusion in a repository or data publication. DataCite includes more fields than DublinCore, such as versioning, publisher, and funding sources. Finally, a custom JSON schema could easily be created to capture the finance specific metadata, such as ticker symbol, exchange, and currency. However, using Dublin Core or DataCite would be more efficient to cite, reproduce, and share the data.

Q4:

The provenance data I can collect for the dataset would start with the raw CSV file exported from Google Forms. I would also include information on creating the survey, how it was distributed, when responses were recorded, sample size, and other activities performed with the dataset such as downloading, cleaning, or transforming the data. The provider, collector, and analyst of the data would be me since I own and collected the data via Google forms. By representing the provenance in terms from W3C PROV-O, I can better describe how different aspects of the workflow interact, creating a more reproducible product. The information will allow others to replicate my steps for data analysis and show transparency on the collection methods.

Answers for Collection B: [The Amazon Daily Stock Dataset](#)**Q1a:**

The amazon daily stock dataset provides historical financial data gathered from Yahoo Finance via the finance python package. The data was collected through different measurement processes at stock exchanges. They recorded daily information on trades and compiled it into the following fields: date (specific trade date), open (opening price), high (maximum price during the day), low (minimum price during the day), close (close price adjusted for splits), adj close (adjusted close price adjusted for both dividends and splits), and volume (the number of shares that changed hands during a given day). I was driven to choose this dataset because I have always wanted to do a data project on finance but have never had the chance. The research idea here is to study trends in stock price fluctuations or trading volumes and find whether large shifts correspond to major company or market events. Additionally, since it is likely that this will

be the case, I would like to connect the specific events to these changes if possible. I can access this dataset for free through kaggle, which makes the project feasible since I only need an internet to download the data one time. The dataset includes information on the years of 1997-2020, which is large enough to provide a meaningful analysis.

Q1b:

- **Creation of logical Collections**

The logical collection for this project will consist of the amazon daily stock trading dataset, organized as a single dataset with consistent structure. This structure contains 7 columns: date, open, high, low, close, adjusted close, volume). Additionally, the data can be separated into subsets (by year, volume, etc.) including processed data such as market volatilities or monthly/quarterly averages to create other logical groupings. All collections would have unique names following a similar naming convention to clearly distinguish different sets of data and help distinguish between the raw and derived data.

- **Physical data handling**

The original data was sourced from Yahoo Finance thanks to the yfinance python package, and the raw CSV file was downloaded from kaggle onto my laptop. A copy will be uploaded to either Google Drive or a github repository for this class to ensure there is a secondary backup. Since I have no reason to edit the original file, it will reduce the risk of any file corruptions or data alterations. As stated before, naming conventions with different folder hierarchies will be used to keep the data organized and locatable. Should the dataset need to be transferred into a physical form, I will use a printer.

- **Interoperability Support**

Since the dataset is already in a CSV format, it will be supported on many platforms such as R, Python, and Excel. If need be, I could convert it into a xlsx if for some reason the CSV is not very readable to the human eye. Should this dataset need to be shared with collaborators, I would show all versions of the file to ensure maximum compatibility. The metadata would be stored in a plain text README file, allowing it to be read on any system. Each of these steps is essential to ensure that the data can be reused and read in different environments without compatibility issues.

- **Security Support**

This dataset does not contain any sensitive personal information, so the only security concerns would be unauthorized modifications or accidental loss. The files will be stored in password protected accounts on my laptop, google drive and/or github. Permissions for any cloud storage would be set to "private" unless explicitly shared with a collaborator. Finally, each time the dataset is open, I will keep it read only to further protect it against any accidental modifications.

- **Data Ownership**

The original owner of the stock dataset is Yahoo Finance, which means I must give them credit everywhere I use the dataset. Any cleaned and processed versions of this data that I use for analysis will be owned by me, as will any subset files. Should I share these files with others, I will always make a distinction between any original files and derivative work that I created, as well as credit the owners of the originals. Data quality can be

ensured by keeping the files read only and comparing them to originals found on Yahoo Finance.

- **Metadata collection, management, and access**

The metadata for the dataset will include information like the source, ticker symbol, dataset title, variables, definitions, and download date. This data will be stored in a README file which will travel with the dataset. Additional metadata will show any steps I took in processing the data, which will be stored in the same folder as the dataset. Keeping this metadata for collaborators to view is imperative to maintaining transparency and correct interpretation of the data.

- **Persistence**

The dataset along with any metadata will be backed up on my github/google drive to maintain long term persistence. The raw dataset shall be preserved without any modifications, and derived datasets will be saved with version numbers in the filenames to differentiate them. Despite the fact that Yahoo Finance or the yfinance python package would allow me to redownload the data, keeping my own copy ensures reproducibility in a scenario where I cannot access either of those tools. Github will ensure that the dataset and the project itself live on past the semester and graduation. Should I want others to access it, I can explicitly share it with them or set the repository to "public".

- **Knowledge and information discovery**

Should this project be shared with others, people would be able to learn from the data without doing their own analysis by looking at the one that I store in the project folder. However, should they choose to do their own analysis they would be able to do so. I would provide discovered trends in stock performance such as volatility periods, patterns, or unusual spikes in trading volume. If I wanted to make it even easier for others to search, filter, or query the dataset, I could store it in an SQL database or provide visualizations like plots or histograms.

- **Data distribution and publication**

The dataset is public, which means I can freely share any processed or raw versions of this data. If I were to publish my processed dataset through a means such as a github repository, I would provide my processed data, the original data, and a README file with the metadata and a link to the page where I downloaded the dataset. I would also make it clear that I do not own the original data and only used it for the purposes of my project, showcasing the results as well. Any advertisements of this project (such as including it on a resume) would note that I had used publicly available stock data from Yahoo Finance for my project.

Q2:

The stock dataset was originally provided in a CSV, which is typical for large datasets such as this due to how many platforms support it. CSV is also a very readable format for humans wishing to interpret the data. However, it comes with storage limitations and little built in validation. Apache Parquet is a good alternative if need be, as it uses columnar storage. Should I decide to combine this dataset with those of other stocks, this format would be extremely useful. JSON is also a suitable format, as it allows for structured nesting and visible metadata.

Should the dataset need to be viewed from an SQL based platform, it could be converted to SQL for easy queries and joining of other data. Finally, the excel file type, XLSX, would be useful for smaller analysis or sharing information in a business context since it is typically used by less technical users. The downside for this choice is that it can be a hassle to reproduce large scale work. Overall, CSV is the most practical choice due to its large support network, but Parquet and SQL would be suitable for a long term project.

Q3:

There are a few metadata standards that could be useful for this dataset. Dublin Core is widely used for dataset description, including fields such as title, creator, source, subject, and date. This schema is particularly good for inclusion in a repository or data publication. DataCite includes more fields than DublinCore, such as versioning, publisher, and funding sources. Finally, a custom JSON schema could easily be created to capture the finance specific metadata, such as ticker symbol, exchange, and currency. However, using Dublin Core or DataCite would be more efficient to cite, reproduce, and share the data.

Q4:

The provenance data I can collect for the dataset would start with the raw CSV file, which would provide me the URL, ticker symbol, download date, and file checksum. I would also include information on activities performed with the dataset, such as downloading, cleaning, or transforming the data for my needs. The provider of the data would be Yahoo Finance by proxy of the yfinance python package and I would be the collector/analyst. By representing the provenance in terms from W3C PROV-O, I can better describe how different aspects of the workflow interact, creating a more reproducible product. The information also allows anyone to trace the dataset back to its origin and replicate my steps.